Closed-form potentials for image denoising

From marginals to MMSE-matched MAP priors — no training

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The question, in one slide

A **fields-of-experts** (FoE / ridge) regularizer scores an image by passing it through filters and adding up a 1-D *potential* per filter response:

$$R(x) = \sum_{k=1}^{K} \sum_{i} \lambda_k\big((h_k * x)_i\big).$$

Denoising = MAP estimation:

$$\hat{x} = \arg\min_x \ \tfrac{1}{2}\|x - y\|^2 + \beta R(x).$$

WCRR (Goujon, Neumayer, Unser 2024) **trains** all of it end-to-end ($\approx 3\,\mathrm{h}$).

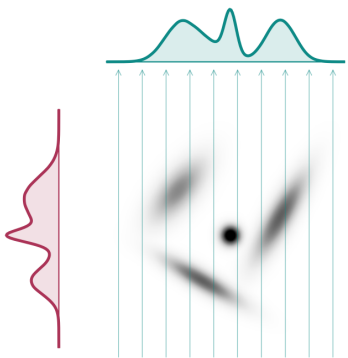
Question. Can the potentials be **computed** — from image statistics, by explicit formulas — and still be as good as trained ones?

Answer (same 60 filters as WCRR, $\sigma = 25$)

	Set12	BSD68
WCRR (**trained**)	29.618	28.681
ours, unfiltered	28.985	27.873
ours, filtered, $\beta = \sigma^2$	**29.742**	**28.712**

Potentials: **3 linear solves**, $\approx 3\,\mathrm{min}$ (WCRR: 3 h).
β: **predicted** by theory, not tuned.

The starting idea: a filter histogram is a *projection* of the image density known



a density in $\mathbb{R}^2$ (grey) and two of its projections: integrate along the rays, record the profile

An image is a point $x \in \mathbb{R}^N$; natural images have a density $p(x)$. A filter response $z = \langle h, x \rangle$ *projects* x onto the direction h. Its histogram is

$$\rho_h(t) = \int p(x)\, \delta\big(t - \langle h, x \rangle\big)\, dx \;=\; (\mathcal{R}p)(h, t),$$

the **Radon transform** of the density along h.

- All projections together determine p (Cramér–Wold).
- A filter bank $\times$ all shifts = many projection directions; their histograms are cheap to measure.
- **The tomography question**: rebuild the density — or rather its potential $-\log p$ — from its projections.

Maximum entropy turns projections into a fields of experts

known

Among all densities with the measured projections, take the one with maximum entropy:

$$\max_{q} \; H(q) \quad \text{s.t.} \quad (\mathcal{R}q)(h_k, \cdot) = \rho_k \quad \forall k.$$

The Lagrange multipliers are whole functions φ_k, and the solution is

$$q(x) \;\propto\; \exp\Big(\sum_k \varphi_k\big(\langle h_k, x\rangle\big)\Big)$$

and with convolution filters (all shifts): $\log q(x) = \sum_{k,i} \varphi_k\big((h_k * x)_i\big) + \text{const}$.

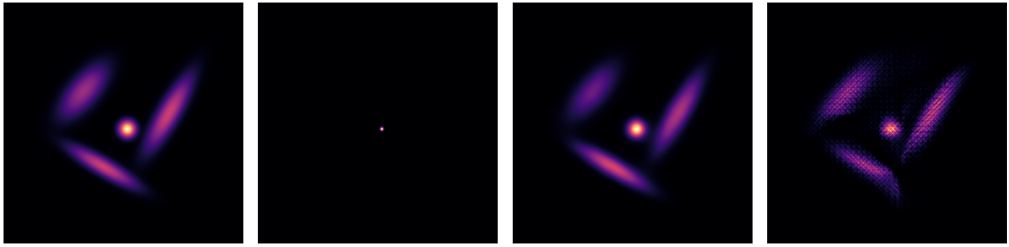
That is exactly a **fields-of-experts** prior, with potentials $\lambda_k = -\varphi_k$.

Back-projection. In tomography, $(\mathcal{R}^* g)(x) = \sum_k g_k(\langle h_k, x\rangle)$: a sum of ridge functions. So the log-density is the *back-projection* of the φ_k.

Which φ_k? The naive choice back-projects the log-histograms: $\varphi_k = \log \rho_k$ (product of marginals).

FRAME (Zhu, Wu, Mumford 1998): max-entropy with filter histograms, fitted by Gibbs sampling. Exact fit: iterative proportional fitting (Deming–Stephan 1940). Projection pursuit density estimation (Friedman et al. 1984).

Unfiltered back-projection over-counts; one linear solve fixes it (2-D testbed)



target density

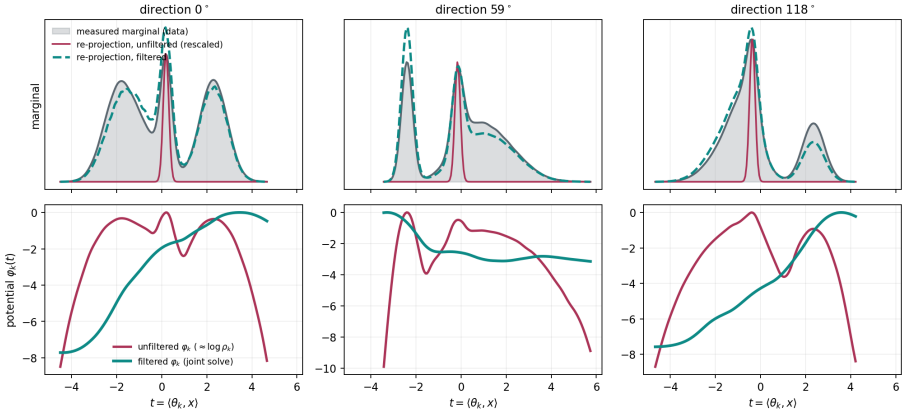
unfiltered: each direction alone
KL = 39.5

filtered: one linear solve
$Ac = d$
KL = 0.03

iterative max-entropy (IPF, 2 sweeps)
KL = 0.25

64 directions in $\mathbb{R}^2$, so every direction overlaps its neighbours (redundancy $\kappa = K/N = 32$). Model: $\log q(x) = \sum_k \varphi_k(\langle \theta_k, x \rangle)$, fitted by score matching — the same normal equations as for images. **Unfiltered** = keep only the diagonal blocks of A (each direction fitted alone, $\approx$ log-histograms): information counted $\approx \kappa$ times, the model collapses. **Filtered** = the full solve: the overlap is undone and the density is recovered.

Testbed: marginals and potentials



Top (tomographic check): projecting each fitted model back onto a direction. The filtered model reproduces the measured marginal; the unfiltered one collapses to a spike.

Bottom: unfiltered potentials are the log-histograms. Filtered potentials are *shares*: smoother, and they trade tilts and curvature with the other directions, so only their sum is meaningful.

Score matching *is* filtered back-projection, in closed form

new here

The normal equations $A\,c = d$ of the next section split exactly like a tomographic reconstruction:

- $d_{(k,a)} = \mathbb{E}\sum_i b'_a(z_{k,i})$ (Stein's lemma) uses **only the histogram of channel** k: the back-projected data.
- $A_{(k,a),(l,b)} = \mathbb{E}\langle h_k^\top * b_a(z_k),\, h_l^\top * b_b(z_l)\rangle$ measures how the projection directions **overlap**: the normal operator, like $\mathcal{R}^*\mathcal{R}$ in CT.
- $c = A^{-1}d$: **filtered** back-projection. Keep only the diagonal blocks of A and you get the per-channel (marginal) potentials: **unfiltered**.

tomography (CT)	image priors (here)
object f	potential $-\log p$ of images
angle θ	filter h_k (and its shifts)
projection $\mathcal{R}_\theta f$	histogram of $h_k * x$
back-projection $\mathcal{R}^* g$	$\sum_{k,i} \varphi_k((h_k * x)_i)$ (FoE)
unfiltered BP: blurred	product of marginals: over-counted ($\beta^\star \approx \beta_{\mathrm{cal}}/230$)
ramp filter $\lvert\omega\rvert$	A^{-1}: undo the overlap
iterative (ART)	IPF: exact max-entropy
FBP, closed form	one linear solve

In images the "ramp filter" is not a fixed $|\omega|$: A is measured from data, so it is the exact correction for this bank and this model class. The rest of the talk: which density to reconstruct (the *noisy* one), and how to turn it into MAP potentials.

Notation (everything else is built from this)

Data

- clean image $x \in [0,1]^N$, noisy $y = x + \sigma n$, $n \sim \mathcal{N}(0, I)$
- $\sigma = 25/255$ throughout
- filters $h_1, \ldots, h_K$: zero mean, $\|h_k\| = 1$; responses $z_k = h_k * x$
- $h_k^\top *$: the adjoint (correlation) of $h_k *$

Densities

- p: density of clean images; $p_\sigma = p * \mathcal{N}(0, \sigma^2 I)$: noisy images
- ρ_k: marginal density of one response $z_{k,i}$

Model class (two kinds of experts)

$$R(x) = \underbrace{\sum_{k,i} \lambda_k(z_{k,i})}_{\text{1-D experts}} + \underbrace{\sum_{k,i} \tfrac{1}{m} F_k(\log A_{k,i})}_{\text{scale field}}$$

$A_{k,i} = \sum_{j \in \text{window}(i)} z_{k,j}^2 / s_k^2$: local energy of filter k in a $(2r+1)^2$ window, $m = (2r+1)^2$.

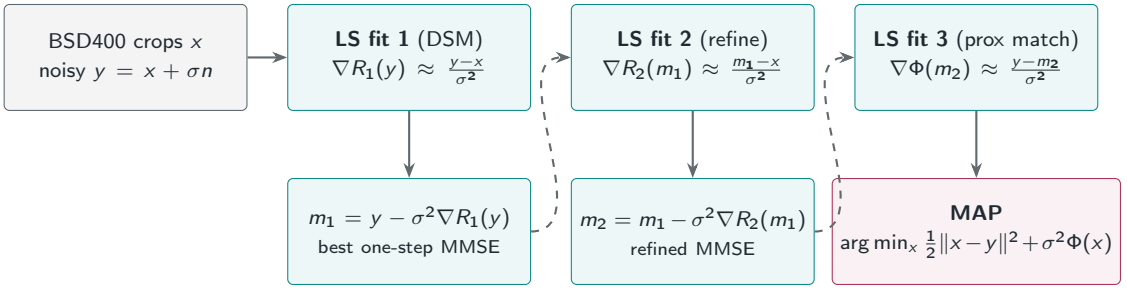
Units. A penalty is drawn in *data-term units*, i.e. as $\beta\lambda_k$: what is added to $\frac{1}{2}\|x - y\|^2$. Then ours and WCRR's are directly comparable.

What is trained, what is explicit

ingredient	WCRR	ours
filters h_k	**trained** (bilevel, deep equilibrium)	*given*: WCRR's own 60 (like-for-like test), or V4 = DCT + steerable pyramid (**analytic**)
1-D potentials λ_k	**trained** spline, weakly convex by design	**explicit**: one linear least-squares solve
scale-field term F_k	—	**explicit**: same linear solve
weight β	**trained** noise-dependent scaling	**predicted** $\beta = \sigma^2$ (or 1 scalar **tuned** on 16 images)
ridge / roughness	(training choices)	**chosen** on 44 held-out images, same objective
data	BSD400 crops	the same BSD400 crops
cost	≈ 3.0 h on a V100 (measured)	2.8 min on the same GPU type, 3 linear solves

teal = closed form **wine** = trained by gradient descent **amber** = chosen on held-out training images. *No test image is ever used.*

The whole method on one page



Every box in teal is **the same routine**: least squares for a gradient field that is *linear in the coefficients*, evaluated at some points p, matched to some target t. Only the pair (p, t) changes.

Fact 1 — Tweedie: the best denoiser is one gradient step

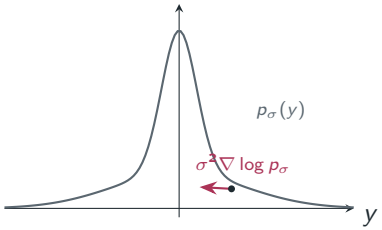
known

For Gaussian noise, the posterior mean (MMSE estimator) is

$$\mathbb{E}[x \mid y] = y + \sigma^2 \nabla \log p_\sigma(y).$$

(Robbins 1956, Miyasawa 1961, Efron 2011.) One line of proof: $\nabla p_\sigma(y) = \int \frac{x-y}{\sigma^2}\, p(y|x)p(x)\, dx$, divide by $p_\sigma(y)$.

Reading. If we knew the *noisy*-image density p_σ, the optimal denoiser would be a single gradient step on $\log p_\sigma$ from y, with step σ^2.



Fact 2 — Denoising score matching is plain regression

known

Choose R to make the one-step denoiser $y - \sigma^2 \nabla R(y)$ as good as possible. Its error is

$$\mathbb{E}\left\| \left(y - \sigma^2 \nabla R(y)\right) - x \right\|^2 = \sigma^4 \, \mathbb{E}\left\| \nabla R(y) - \frac{y - x}{\sigma^2} \right\|^2 .$$

The right-hand side is **denoising score matching** (Vincent 2011): regress the gradient field ∇R on the target $(y - x)/\sigma^2$ over pairs (x, y). Its minimiser over all functions is $\nabla R = -\nabla \log p_\sigma$, i.e. Tweedie.

Two consequences we use

1. Fitting R by DSM *at the noise level of the task* gives the best one-step denoiser in the model class — **no step size to tune** (it must be exactly σ^2).
2. The held-out DSM objective J_{val} *is* the held-out MSE: $\text{MSE}_{\text{out}} = \text{MSE}_{\text{in}} + 2\sigma^4 J_{\text{val}}$. Model selection on J_{val} = selection on held-out PSNR.

Fact 3 — The MMSE denoiser is itself a MAP estimator

known

$D(y) = \mathbb{E}[x|y] = \nabla\psi(y)$ with
$\psi(y) = \frac{1}{2}\|y\|^2 + \sigma^2 \log p_\sigma(y)$, and

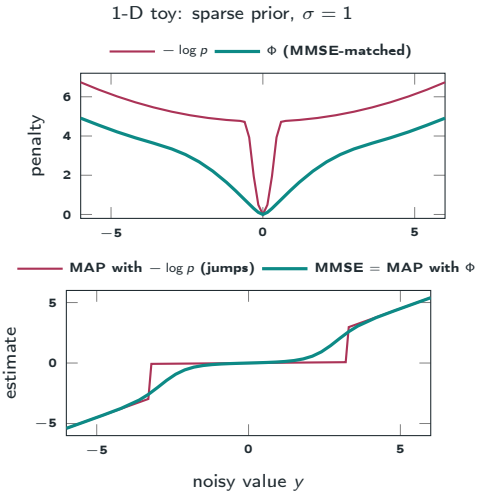
$$\nabla^2\psi(y) = \mathrm{Cov}[x \mid y]/\sigma^2 \succeq 0.$$

So D is the gradient of a *convex* function. Gribonval (2011) and Gribonval–Nikolova (2020): then D is a proximal operator,

$$D(y) = \arg\min_x \ \tfrac{1}{2}\|x - y\|^2 + \sigma^2\Phi(x),$$

with $\sigma^2\Phi + \frac{1}{2}\|\cdot\|^2$ convex (**weakly convex** Φ).

Reading. MAP with the true prior $-\log p$ is *not* optimal. MAP with Φ is. $\Phi \neq -\log p$: it is smoother. Its weak convexity is exactly the constraint WCRR builds in.



Linear in the coefficients ⇒ one linear solve

Expand each potential's derivative on a fixed basis (hat functions on log-spaced knots):

$$\lambda'_k(z) = \psi_k(z) = \operatorname{sign}(z) \sum_{a=1}^{29} c_{k,a}\, b_a\big(|z|/c_k\big), \qquad F'_k(L) = \sum_{a=1}^{16} e_{k,a}\, g_a(L).$$

Then the gradient of the whole energy is a **linear combination of fixed images**:

$$\nabla R(p) = \sum_j c_j\, G_j(p), \qquad G_{(k,a)}(p) = h_k^{\top} * \big[b_a(h_k * p)\big] \quad \text{(and similarly for the scale field).}$$

Normal equations

$$\min_c \; \mathbb{E}\Big\| \sum_j c_j G_j(p) - t \Big\|^2 \iff \underbrace{\mathbb{E}\langle G_i(p), G_j(p)\rangle}_{A_{ij}}\, c_j = \underbrace{\mathbb{E}\langle G_i(p), t\rangle}_{d_i}$$

60 filters: $D = 60 \times (29 + 16) = 2700$ unknowns; A accumulated over 340 crops (≈ 8.8 M pixels) on the GPU in float64, then $(A + \text{ridge} + \text{roughness})\, c = d$.

The three fits are one routine with three (points, target) pairs

fit	points p	target t	what it gives
1. DSM step	y	$\dfrac{y-x}{\sigma^2}$	$R_1 \approx -\log p_\sigma$; $m_1 = y - \sigma^2 \nabla R_1(y)$ (Tweedie)
2. refinement step	m_1	$\dfrac{m_1-x}{\sigma^2}$	$m_2 = m_1 - \sigma^2 \nabla R_2(m_1)$ (better MMSE estimate)
3. prox matching	m_2	$\dfrac{y-m_2}{\sigma^2}$	Φ with $\mathrm{prox}_{\sigma^2\Phi}(y) \approx m_2$: **the MAP potentials**

Why fit 3 is the right target

MAP optimality of $\hat{x}$: $y - \hat{x} = \sigma^2 \nabla \Phi(\hat{x})$. We want $\hat{x} =$ MMSE estimate m_2. So $\nabla\Phi$ must map $m_2 \mapsto (y - m_2)/\sigma^2$: a regression, linear in Φ's coefficients.

Consequence

MAP with Φ is run at $\beta = \sigma^2 = \beta_{\mathrm{cal}}$, the Bayesian weight. **Nothing to tune**. Measured optimum on held-out images: $1.06\,\beta_{\mathrm{cal}}$.

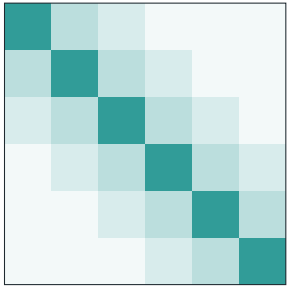
Unfiltered vs filtered on images: who owns the evidence?

Unfiltered (marginals): $\lambda_k = -\log \rho_k$, each filter alone. Filters overlap: they see the same pixels. Each one claims the full evidence, so it is counted $\approx$ many times.

Filtered: solve the *joint* normal equations. The off-diagonal blocks of A measure how much the gradient fields of two filters overlap; solving $Ac = d$ shares the evidence among them, like multiple regression with correlated regressors.

Over-counting, measured: the unfiltered prior needs $\beta^\star = 0.0044\,\beta_{\mathrm{cal}}$ ($\approx 1/230$). The filtered one: $\beta^\star = 1.06\,\beta_{\mathrm{cal}}$.

$A = [\mathbb{E}\langle G_i, G_j\rangle]$

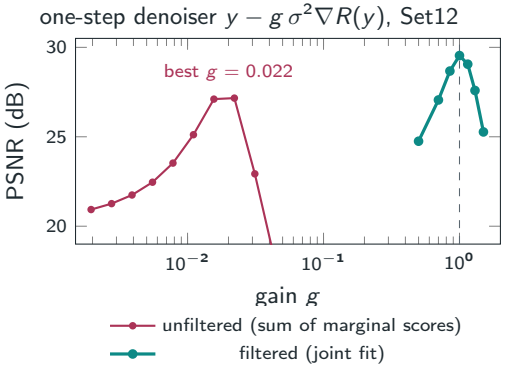


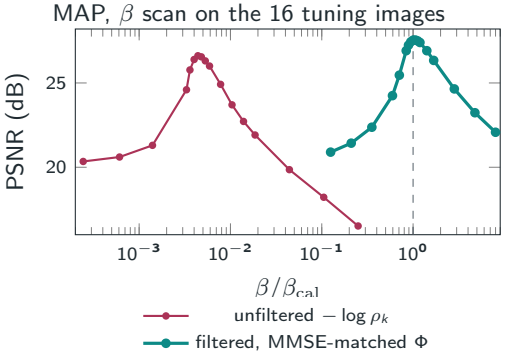
diagonal blocks: one filter alone (marginals)

off-diagonal: overlap between filters

Unfiltered $\approx$ keeping only the diagonal.
Filtered = the full matrix.

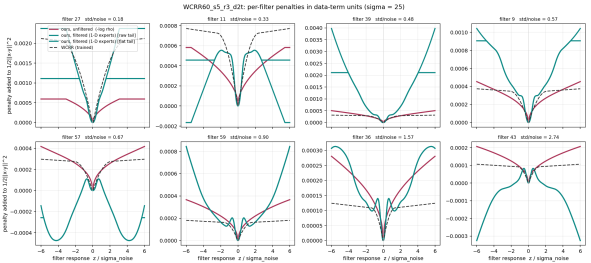
Calibration: no gain, no β to tune — only when filtered





The filtered models peak **exactly where theory says** ($g = 1$, $\beta = \beta_{\mathrm{cal}}$). The unfiltered ones need a factor $\approx 1/45$ (one step) and $\approx 1/230$ (MAP), and still lose > 0.6 dB.

Redundant filters ⇒ per-filter potentials are not identifiable



Before: unconstrained fit (earlier pipeline). Wiggles, dips, even negative tails.

Many combinations of per-filter potentials give almost the same *total* gradient (A has a near-null space). The unconstrained optimum picks an arbitrary one.

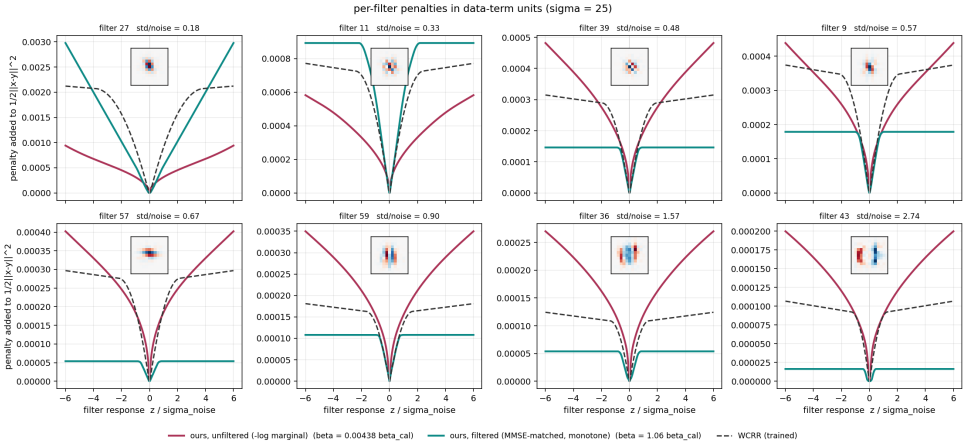
Shape constraint (convex)

$c_{k,a} \geq 0$ for all 1-D coefficients $\iff$ each λ_k is even and **non-decreasing in** $|z|$ (WCRR's class).

Box QP $\min_{c \geq 0} \frac{1}{2} c^\top Q c - d^\top c$: ADMM (one Cholesky) to find the active set, then a primal–dual active-set method to the exact KKT point.

Costs 0.3% of the held-out objective; MAP on Set12: **29.17** → **29.82** dB.

The potentials, side by side (WCRR's 60 filters, data-term units)



Unfiltered: concave cusps that keep growing (needs $\beta \approx \beta_{\mathrm{cal}}/230$). Filtered, MMSE-matched: quadratic core of width $\approx \sigma$, flat beyond — *shrink what looks like noise, leave signal alone*. Dashed: WCRR, trained.

Implementation details that matter

Fitting (`afoe.py`, ≈700 lines, PyTorch)

- responses and adjoints by FFT; feature images in chunks of 16 filters
- A, d accumulated in float64 over the interior (10 px border) of 340 crops
- hyper-parameters (ridge, roughness) on 44 *held-out* crops, by J_{val}
- potentials stored as fine tables (96 001 points) with exact cell slopes: energy and gradient are consistent, so line searches are valid
- data split fixed: 340 fit / 44 select / 16 tune (BSD400); tests never touched

MAP solver

- FISTA with Beck–Teboulle backtracking, momentum restart when the energy rises, batched
- bug found: the exact backtracking test fails on float32 rounding near convergence, and one failing image made the whole batch backtrack 40× per iteration (BSD68 78× slower). Fix: compare energies up to 10^{-6} relative.
- converged at 250 iterations (PSNR identical to 4 decimals at 250/500/1000/2000); we use 500
- reflect padding; noise seeded per (set, image, σ): every method sees the same noise

Results (1/2): MAP with explicit potentials on WCRR's own 60 filters

potentials	β	PSNR (dB), $\sigma = 25$ Set12	PSNR (dB), $\sigma = 25$ BSD68	# images > WCRR Set12	# images > WCRR BSD68
WCRR (**trained**)	**trained**	29.618	28.681	—	—
unfiltered $-\log \rho_k$	**tuned**: $0.0044\,\beta_{\text{cal}}$	28.985	27.873	0/12	0/68
filtered, 1-D only (= WCRR's class)	β_{cal}	29.415	28.561	0/12	7/68
filtered, 1-D only	**tuned**: $1.06\,\beta_{\text{cal}}$	29.521	28.603	3/12	9/68
filtered, 1-D + scale field	β_{cal}	**29.742**	**28.712**	11/12	36/68
filtered, 1-D + scale field	**tuned**: $1.06\,\beta_{\text{cal}}$	**29.808**	**28.737**	12/12	43/68

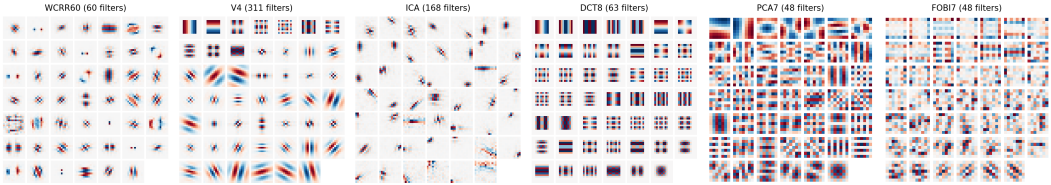
Same filters, same data, same noise realisations; WCRR re-run with the authors' model and solver. Only the potentials differ: **trained** vs **three linear solves**.

Results (2/2): explicit estimators without MAP, and other banks

		PSNR (dB), $\sigma = 25$		# images > WCRR	
method	filters	Set12	BSD68	Set12	BSD68
BM3D (reference)	—	29.984	28.616	10/12	18/68
WCRR (**trained**)	60 **trained**	29.618	28.681	—	—
one closed-form step	WCRR's 60	29.536	28.576		
two closed-form steps	WCRR's 60	29.770	28.702	10/12	34/68
two closed-form steps	V4 (**analytic**, 311)	29.664	28.658	4/12	25/68
two closed-form steps	WCRR60 + V4 (371)	**29.922**	**28.818**	12/12	61/68

No β, no step size, no tuning: each step is $m \leftarrow m - \sigma^2 \nabla R(m)$ with the fitted R. V4 = DCT-8 + steerable pyramids: a filter bank that involves no learning at all.

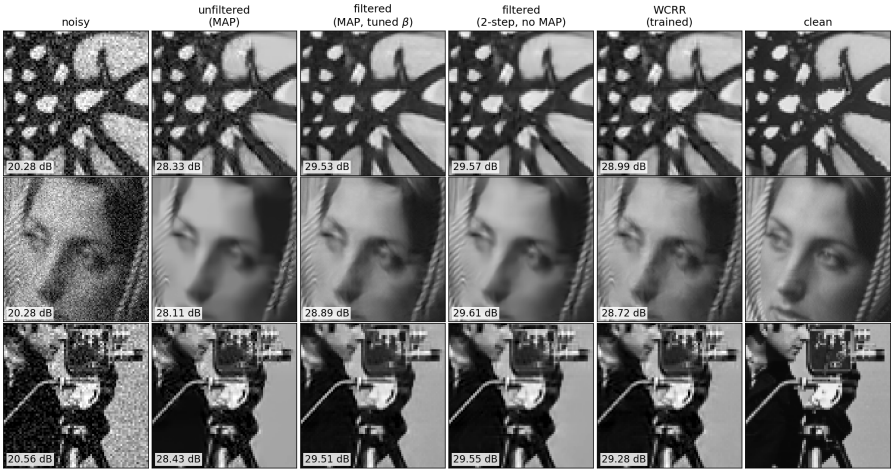
Which filters? Learned, analytic, and data-analytic



filters	obtained by	K	held-out (VAL)	Set12	BSD68
WCRR60 + V4 ($r = 5$)	learned + analytic	371	28.05	29.92	28.82
WCRR60	**learned** (bilevel training)	60	27.93	29.77	28.70
V4	**analytic**: DCT-8 + steerable pyramids	311	27.84	29.66	28.63
ICA 13×13	data: FastICA (iterative)	168	27.81	29.53	28.57
PCA 5×5	data: patch PCA, **closed form**	24	27.60	29.25	28.41
DCT 8×8	**analytic**	63	27.59	29.32	28.40
PCA 13×13	data: patch PCA, **closed form**	168	27.44	29.12	28.27
FOBI 5×5	data: closed-form ICA	24	27.39	28.90	28.14

Two explicit steps, $r = 3$, no tuning. **The held-out objective ranks the banks exactly like the BSD68 test PSNR**: it is the explicit criterion for choosing filters. PCA is optimal only for the Gaussian part; the experts need localized, oriented filters.

What it looks like (Set12: Monarch, Barbara, Cameraman; zooms)



PSNR on the full image. Unfiltered: blotchy noise, erased stripes (over-weighted prior). Filtered: clean and sharp, texture kept.

Time benchmark (one Tesla V100-SXM2-32GB, same GPU for every row)

Getting the potentials (from scratch, incl. data loading and selection)

LS fit 1 (DSM step)	59.5 s
LS fit 2 (refinement step)	47.3 s
LS fit 3 (prox matching)	60.0 s
ours, total	**2.8 min**
WCRR **training**	**3.03 h**

WCRR: the repository's own training, measured earlier on a V100: 1.82 s/batch × 6000 batches. Ratio ≈ 65×.

Denoising one image (median of 3 runs, after warm-up)

	481×321	512×512
ours: two closed-form steps	**0.015 s**	**0.022 s**
ours: MAP, 1-D + SF, 500 it	11.7 s	17.9 s
ours: MAP, 1-D + SF, 250 it*	6.0 s	9.1 s
ours: MAP, 1-D only, 500 it	5.5 s	7.8 s
unfiltered MAP, 4000 it	41.7 s	59.5 s
WCRR, its own solver	1.5 s	2.2 s

*already converged (same PSNR to 4 decimals as 500–2000 iterations).

Fitting is ≈ 65× cheaper than training. The explicit two-step denoiser is 100× faster than WCRR. Our MAP solver is 4–8× *slower* than WCRR's (generic FISTA with line search on tabulated potentials): inference speed is the open engineering item.

How to read the results

1. **Filtering = sharing evidence.** Marginal potentials are right for one filter alone; with overlapping filters, the joint least squares divides the evidence. The measured over-counting factor ($\beta^\star \approx \beta_{\mathrm{cal}}/230$) is the price of ignoring it.
2. **The best MAP prior is not $-\log p$.** The MAP-optimal penalty is the MMSE-matched Φ: quadratic up to about the noise level, then saturating. MAP with $-\log p$ thresholds hard (the cliff in the β scan); MAP with Φ shrinks smoothly.
3. **WCRR's design is a theorem.** Weak convexity and the saturating shape, imposed in WCRR by architecture, are properties of the exact MMSE-matched penalty.
4. **β is predicted.** $\beta = \sigma^2$ follows from Tweedie; the held-out optimum is $1.06\,\beta_{\mathrm{cal}}$, and using β_{cal} instead costs 0.03–0.11 dB.
5. **A local-variance term is what beats WCRR.** In WCRR's exact class (1-D experts, same filters) the explicit potentials are 0.08–0.10 dB below; adding the explicit scale field puts them 0.03–0.19 dB above (both sets, with or without tuning).

What is *not* new (we build on it)

Known results used here

- Radon transform, Cramér–Wold; maximum entropy with filter histograms (FRAME, Zhu–Wu–Mumford 1998); iterative proportional fitting; projection pursuit density estimation.
- FoE / ridge regularizers: Roth & Black 2005; Chen & Pock 2017; WCRR (Goujon et al. 2024).
- Tweedie / Miyasawa formula; score matching (Hyvärinen 2005); denoising score matching (Vincent 2011).
- **Jointly optimal, linear-in-parameter shrinkage in redundant representations beats per-subband (marginal) shrinkage**: Raphan & Simoncelli 2008 (SURE, per image), Hel-Or & Shaked 2008 (supervised). *Our one-step "filtered vs unfiltered" result is theirs.*
- Gaussian scale mixtures / local-variance models: Wainwright & Simoncelli 2000; Portilla et al. 2003 (and Raphan–Simoncelli's neighbourhood estimators).
- Cascades of shrinkage stages: Schmidt & Roth 2014 (shrinkage fields).
- The MMSE denoiser is the prox of a weakly convex penalty: Gribonval 2011; Gribonval & Nikolova 2020; used for plug-and-play convergence by Hurault et al. 2022 (implicit penalty, trained network).

What is new here (to be checked against a full literature search)

Methodology

- **Prox matching as linear least squares**: explicit FoE potentials whose MAP estimate reproduces a (closed-form) MMSE estimate — a MAP regularizer computed, not trained.
- Reading score matching as **filtered back-projection in log-density space**: the data term is the back-projected histograms, the Gram matrix is the ramp filter (unfiltered = drop its off-diagonal).
- One routine for everything: DSM, refinement and prox matching differ only by (points, target); the held-out objective equals the held-out MSE.
- A monotone shape constraint (exact box QP) that removes the non-identifiability of potentials in redundant banks; it is essential for MAP ($29.17 \to 29.82$ dB).

Results and interpretation

- $\beta = \sigma^2$ *predicted* for MAP and confirmed (held-out optimum $1.06\,\sigma^2$).
- On WCRR's own filters, explicit potentials (+ explicit scale field) **beat the trained potentials on Set12 and BSD68**, without tuning.

Limits and next steps

- **1-D experts alone** (exactly WCRR's class) stay 0.08–0.10 dB below WCRR: the win needs the scale-field term.
- **Large banks** (V4, union): the fitted scale-field terms *decrease* with local energy on 31–48% of their range (they reward contrast), and the projected MAP becomes unstable on high-contrast images (V4: 28.9 dB). Forcing every term to be non-decreasing repairs V4 (29.42) but costs 0.26 dB on WCRR's filters. Open.
- Only $\sigma = 25$ and denoising so far; the next test is other σ and inverse problems (deblurring, MRI), where an explicit regularizer is the point.
- BM3D is still ahead on Set12 (the two-step union estimator: -0.06 dB); WCRR's own advantage over BM3D is on BSD68, where we now beat both.
- Novelty claims need a careful literature search, especially around Hurault et al. 2022 and learned proximal networks.

Backup: what did not work (and why it was informative)

- **DSM at a small noise level, then MAP with tuned** β (the old route): best 29.37 Set12. σ_{fit} sweep 3/4/5/7/10 $\rightarrow$ 29.25/29.34/29.37/29.13/28.57. Starting MAP from y, from the clean image, or from WCRR's output gives the same PSNR: *the energy*, not the optimiser, was the limit.
- **Per-channel Gribonval transform** of the filtered potentials: 25.1 dB. The MMSE-to-MAP conversion must be done jointly (prox matching), not filter by filter.
- **Unconstrained prox matching**: 29.17 dB (vs 29.82 monotone).
- **Flat tails** beyond the fitted range: collapse (22.0 dB with the scale field).
- **More data / knots** (8 flips, 40 knots): no change; larger scale-field window: $r = 5$ best for the one-step estimator ($+0.035$ dB on BSD68 over $r = 3$).

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